Discrete Planning

Vikas Dhiman

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1 Required readings

- Ch 7 and 23, Carmen's Intro to Algorithms
- Breadth-first search and its uses (article) | Khan Academy
- PythonRobotics/dijkstra.py
- Discrete Planning by Lavalle
- 3.5.2 of Russel and Norvig: Artificial Intelligence

sideways ← t+2

t rot move

t+1

State = ? | State space

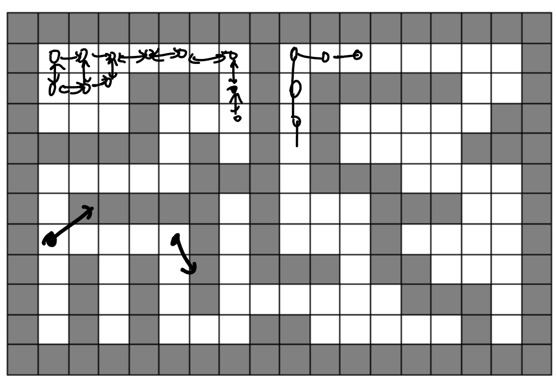
↳ a) Time varying property

2 Discrete planning in Robotics

For robot, that cannot move sideways and have to turn

state = (x, y, θ)

$(-\pi, \pi)$



y

x x

b) Future action should be completely described by state

↳ $a_t \perp s_{t-1} \mid s_t$

Markovian

2.0.1 Abstraction of a planning problem

1. State space $\mathbf{s} \in \mathcal{S}$. For example, 2D coordinate of a grid $\mathbf{s} = (x, y)$.
2. Action space per state $\mathbf{u} \in \mathcal{U}(\mathbf{s})$. For example, up, down, left right movement can be encoded as $\mathcal{U}(\mathbf{s}_t) = \{(0, -1), (0, 1), (1, 0), (-1, 0)\}$.

x, y, z 3D , θ, ϕ, ψ roll/pitch/yaw

Euler angles

system dynamics

$\mathbb{P}(\underline{s}_{t+1} | \underline{s}_t, \underline{u}_t) = ?$

state transistion probabilities

$\underline{s}_{t+1} = f(\underline{s}_t, \underline{u}_t)$

next state

state action

$\boxed{s_t} \rightarrow s_{t+1}?$

3. State transition function $\mathbf{s}_{t+1} = f(\mathbf{s}_t, \mathbf{u}_t)$. For example, the up-down-left-right action can be combined as addition to get the next state $\mathbf{s}_{t+1} = \mathbf{s}_t + \mathbf{u}_t$.

4. Initial State $\mathbf{s}_I \in \mathcal{S}$

5. Goal states $\mathbf{s}_G \subseteq \mathcal{S}$

$\underline{s}_t = \begin{bmatrix} x_t \\ y_t \end{bmatrix}$

$\underline{u}_t = \begin{bmatrix} \Delta x_t \\ \Delta y_t \end{bmatrix}$

$U = \left\{ \begin{bmatrix} +1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ +1 \end{bmatrix} \right\}$

right left down up

① State Space : all possible states
② Action space : all possible action
③ $\underline{s}_{t+1} = f(s_t, u_t)$
④ $\underline{s}_0$ ⑤ Goal state $\underline{s}_g$

2.0.2 A Graph

A graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ is defined by a set of vertices $\mathcal{V}$ and a set of edges $\mathcal{E}$ such that each edge $e \in \mathcal{E}$ is formed by a pair of start and end vertices $e = (v_s, v_e), v_s \in \mathcal{V}, v_e \in \mathcal{V}$. The first vertex is called the start of the edge $v_s = \text{start}(e)$ and second vertex is called the end $v_e = \text{end}(e)$.

A discrete planning problem can be converted into a graph by definiting

$\underline{s}_{t+1} = \underbrace{\underline{s}_t + \underline{u}_t}_{f(\underline{s}_t, \underline{u}_t)}$

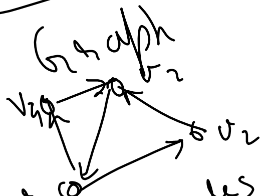
1. Vertices as the state space $\mathcal{V} = \mathcal{S}$. $\in$

2. The action space at each state as the edges connected to that vertex/state, $\mathcal{U}(\mathbf{s}_t) = \{(\mathbf{s}_t, \mathbf{s}_j) \mid (\mathbf{s}_t, \mathbf{s}_j) \in \mathcal{E}\}$ in such that set of edge

3. State transition function is the other end of th edge, $\mathbf{s}_{t+1} = f(\mathbf{s}_t, \mathbf{u}_t) = \text{end}(\mathbf{u}_t)$, where $\mathbf{s}_t = \text{start}(\mathbf{u}_t)$.

$\underline{s}_{t+1} = \begin{cases} \begin{bmatrix} x+1 \\ y \end{bmatrix} & \text{if } u_t = \text{"R"} \\ \begin{bmatrix} x-1 \\ y \end{bmatrix} & \text{if } u_t = \text{"L"} \\ \vdots \end{cases}$

Graph



Vertices/Nodes

Edges that connect them (v_1, v_2) start end

$v_i \equiv s_i$

$\xrightarrow{u_1}$ (s_i, s_j)

2.0.3 Representations of Graphs

(Chapter 23 of Introduction to Algorithms by Carmen et al)

① Adjacency list
② Adjacency Matrix
③ Edge list

Discrete planning → Graph → Graph traversal problem

null ptr

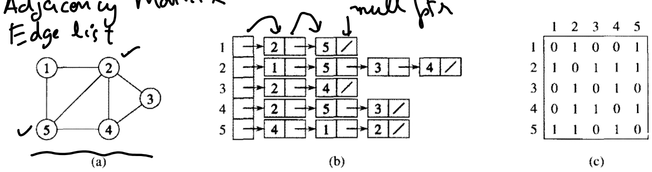


Figure 23.1 Two representations of an undirected graph. **(a)** An undirected graph G having five vertices and seven edges. **(b)** An adjacency-list representation of G. **(c)** The adjacency-matrix representation of G.

Undirected graph

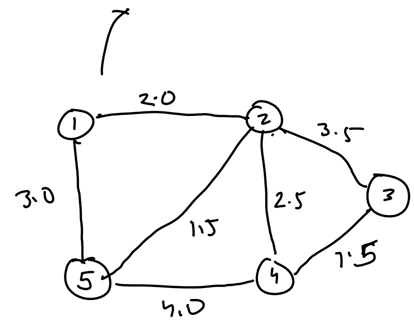
```
# Programmatically you can represent a adjacency list as python lists
# Python lists are not linked lists, they are arrays under the hood.
G_adjacency_list = {
    1 : [2, 5], # node : list of neighbors
    2 : [1, 5, 3, 4],
    3 : [2, 4],
    4 : [2, 5, 3],
    5 : [4, 1, 2]
}

# Prefer to represent a matrix in python either as a list of lists or a numpy array
```

key → (node) ... list of neighbors as values

{
1 : [(2, 2.0), (5, 3.0)] ,
2 : [(1, 2.0), (3, 3.5), (4, 2.5), (5, 1.5)],
⋮ → $O(|E|/|V|)$
}

(2, 4)?
$O(1)$

$$
\begin{array}{c|ccccc}
 & 1 & 2 & 3 & 4 & 5 \\
\hline
1 & 0 & 2.0 & 0 & 0 & 3.0 \\
2 & & & & & \\
3 & & & & & \\
4 & & & & & \\
5 & & & & &
\end{array}
$$




```
import numpy as np
G_adjacency_matrix = np.array([
    [0, 1, 0, 0, 1],
    [1, 0, 1, 1, 1],
    [0, 1, 0, 1, 0],
    [0, 1, 1, 0, 1],
    [1, 1, 0, 1, 0]
])

# Edge list is another possible representation
G_edge_list = {
    (1, 2), (1, 5),
    (2, 1), (2, 5), (2, 3), (2, 4),
    (3, 2), (3, 4),
    (4, 2), (4, 5), (4, 3),
    (5, 4), (5, 1), (5, 2)
}
```

E =

$O(1)$

{
(1, 2) : 3.0
(1, 5) : 4.5
}

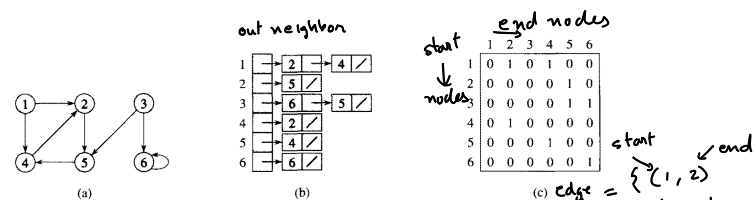


Figure 23.2 Two representations of a directed graph. **(a)** A directed graph G having six vertices and eight edges. **(b)** An adjacency-list representation of G. **(c)** The adjacency-matrix representation of G.

Directed graph representation

```
# Programmatically you can represent a adjacency list as python lists
# Python lists are not linked lists, they are arrays under the hood.
G_adjacency_list = {
    1 : [2, 4],
    2 : [5],
    3 : [6, 5],
    4 : [2],
    5 : [4],
    5 : [6]
}

# Prefer to represent a matrix in python either as a list of lists or a numpy array
import numpy as np
G_adjacency_matrix = np.array([
    [0, 1, 0, 1, 0, 0],
    [0, 0, 0, 0, 1, 0],
    [0, 0, 0, 0, 1, 1],
    [0, 1, 0, 0, 0, 0],
    [0, 0, 0, 1, 0, 0],
```

```
    [0, 0, 0, 0, 0, 1]
])

# Edge list is another possible representation
G_edge_list = [
    (1, 2), (1, 4),
    (2, 5),
    (3, 6), (3, 5),
    (4, 2),
    (5, 6)
]

# Exercise 1

# Write a function that converts a graph in adjacency list format to adjacency matrix and vi
def adjacency_list_to_matrix(G_adj_list):
    G_adj_mat = None # TODO: Write code to convert to adj_mat
    return G_adj_mat

def adjacency_matrix_to_list(G_adj_mat):
    G_adj_list = None # TODO: Write code to convert to adj_mat
    return G_adj_list

# Use the above graphs to test
print(adjacency_list_to_matrix(G_adjacency_list))
print(adjacency_matrix_to_list(G_adjacency_matrix))
```

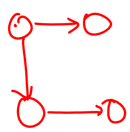
```
None
None
```

Trees don't have cycles/Loops

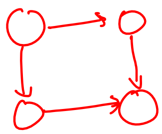
Tree vs

2.0.4 Graph Search algorithms

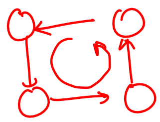
Traversal



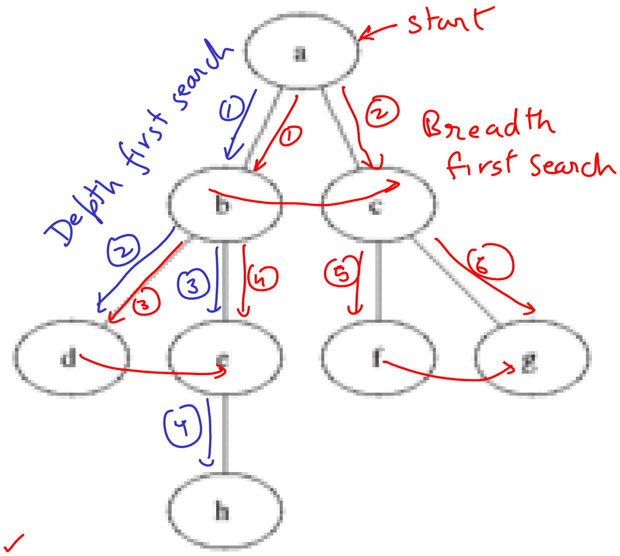
✓ Tree
✓ DAG
✓ Graph

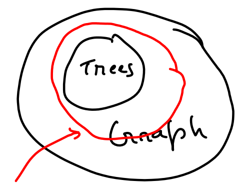


× Not a Tree
✓ DAG
✓ Graph



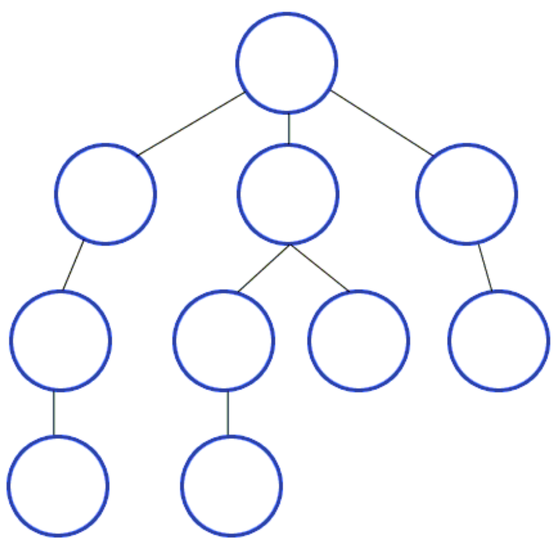
× Tree
× DAG
✓ Graph





DIRECTED
ACYCLIC
GRAPH
(DAG)

1. Breadth First Search ✓
2. Depth First Search ✓



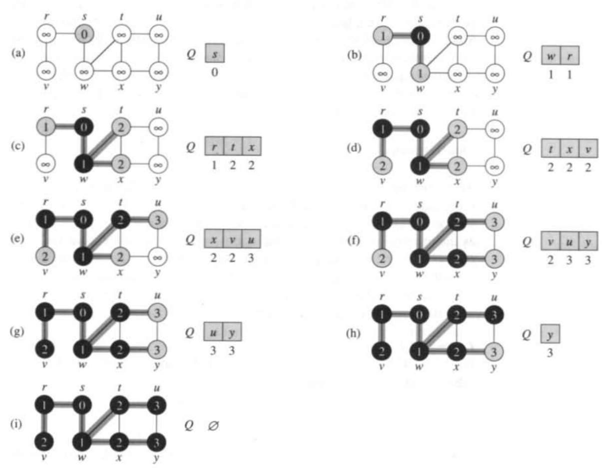


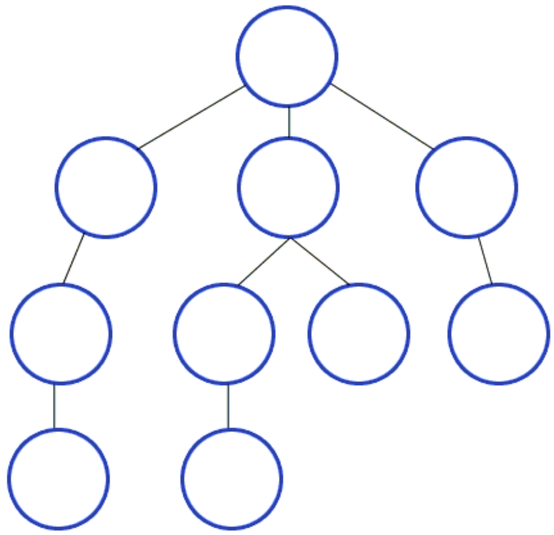
Figure 23.3 The operation of BFS on an undirected graph. Tree edges are shown shaded as they are produced by BFS. Within each vertex u is shown $d[u]$. The queue Q is shown at the beginning of each iteration of the **while** loop of lines 9–18. Vertex distances are shown next to vertices in the queue.

Breadth first search (BFS)

```
from queue import Queue, LifoQueue, PriorityQueue

graph = {
  's' : ['w', 'r'],
  'r' : ['v'],
  'w' : ['t', 'x'],
  'x' : ['y' ],
  't' : ['u'],
  'u' : ['y']
}

def bfs(graph, start, debug=False):
    seen = set() # Set for seen nodes (contains both frontier and dead states)
    # Frontier is the boundary between seen and unseen (Also called the alive states)
    frontier = Queue() # Frontier of unvisited nodes as FIFO
    node2dist = {start : 0} # Keep track of distances
    search_order = []
    seen.add(start)
    frontier.put(start)


    i = 0 # step number
    while not frontier.empty():          # Creating loop to visit each node
        if debug: print("%d) Q = " % i, list(frontier.queue), end='; ')
        if debug: print("dists = " , [node2dist[n] for n in frontier.queue])
        m = frontier.get() # Get the oldest addition to frontier
        search_order.append(m)
```

```
        for neighbor in graph.get(m, []):
            if neighbor not in seen:
                seen.add(neighbor)
                frontier.put(neighbor)
                node2dist[neighbor] = node2dist[m] + 1
            else:
                assert node2dist[neighbor] <= node2dist[m] + 1, 'this should not happen in B
                node2dist[neighbor] = min(node2dist[neighbor], node2dist[m] + 1)

        i += 1
    if debug: print("%d) Q = " % i, list(frontier.queue))
    return search_order, node2dist

print("Following is the Breadth -First Search order")
print(bfs(graph, 's', debug=True))    # function calling
```

```
Following is the Breadth -First Search order
0) Q =  ['s']; dists =  [0]
1) Q =  ['w', 'r']; dists =  [1, 1]
2) Q =  ['r', 't', 'x']; dists =  [1, 2, 2]
3) Q =  ['t', 'x', 'v']; dists =  [2, 2, 2]
4) Q =  ['x', 'v', 'u']; dists =  [2, 2, 3]
5) Q =  ['v', 'u', 'y']; dists =  [2, 3, 3]
6) Q =  ['u', 'y']; dists =  [3, 3]
7) Q =  ['y']; dists =  [3]
8) Q =  []
(['s', 'w', 'r', 't', 'x', 'v', 'u', 'y'], {'s': 0, 'w': 1, 'r': 1, 't': 2, 'x': 2, 'v': 2,
```



BFS

~~Dept~~h first search

gray = frontier
seen = black

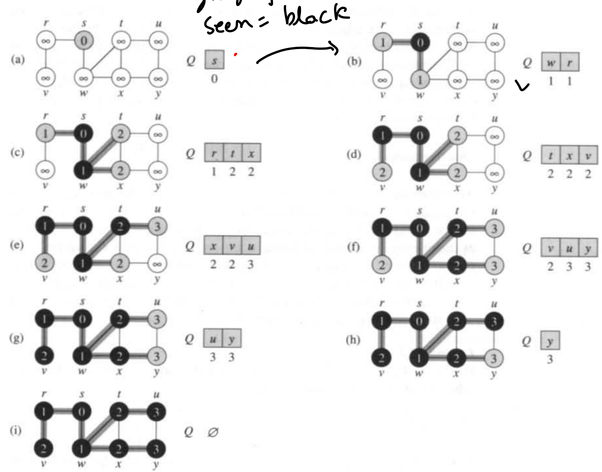
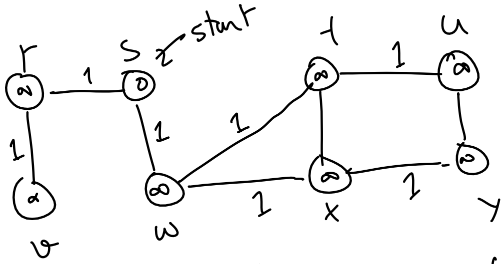


Figure 23.3 The operation of BFS on an undirected graph. Tree edges are shown shaded as they are produced by BFS. Within each vertex u is shown $d[u]$. The queue Q is shown at the beginning of each iteration of the **while** loop of lines 9–18. Vertex distances are shown next to vertices in the queue.

```
graph = {
  's' : ['w', 'r'],
  'r' : ['v'],
  'w' : ['t', 'x'],
  'x' : ['y' ],
  't' : ['u'],
  'u' : ['y']
```

DFS
Depth first Search

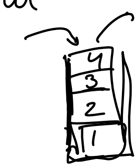


seen = set({})

frontier = Queue() → DFS: FILO stack → BFS: FIFO

nodes that have been visited but their neighbor have not been

FILO : First In Last Out
Stack



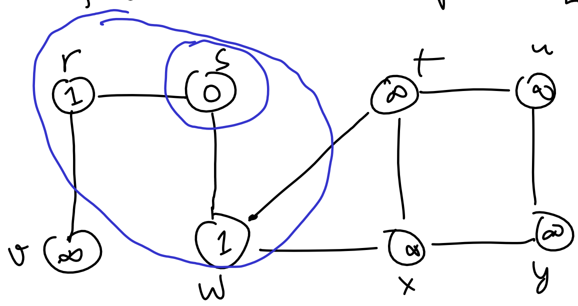
FIFO: First In First Out

BFS

(out)
find the neighbours of s

frontier

s



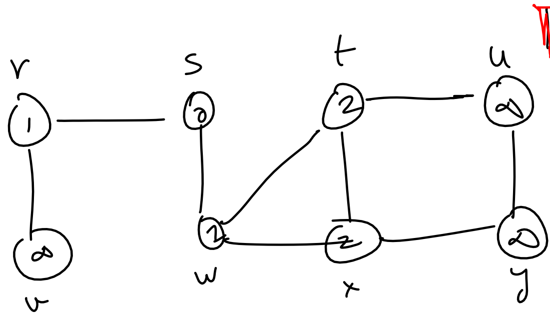
seen

s	r	w

frontier (FILO)

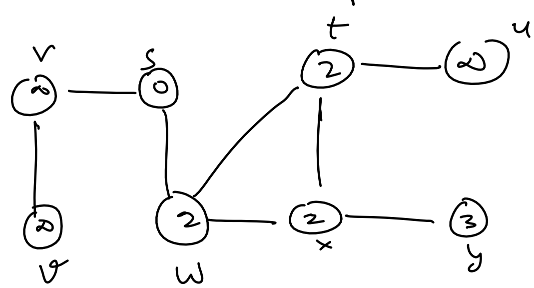
r	w

find neighbors of w



→

find neighbors of x



seen

seen

s	w	t	x	v

frontier

r	t	x

intermediate

seen

s	w	x	t	y	r

frontier

r	t	y

```
}

def dfs(graph, start, debug=False):
    seen = set([start]) # List for seen nodes (contains both frontier and dead states)
    # Frontier is the boundary between seen and unseen (Also called the alive states)
```

frontier = Queue() →

```
    frontier = LifoQueue() # Frontier of unvisited nodes as FIFO
    node2dist = {start : 0} # Keep track of distances
    search_order = [] # Keep track of search order
    frontier.put(start)


    i = 0 # step number
    while not frontier.empty():          # Creating loop to visit each node
        if debug: print("%d) Q = " % i, list(frontier.queue), end='; ')
        if debug: print("dists = " , [node2dist[n] for n in frontier.queue])
        m = frontier.get() # Get the oldest addition to frontier
        search_order.append(m)
```

adjacency list

graph[m] may throw KeyError

```
        for neighbor in graph.get(m, []):
            if neighbor not in seen:
```

default

edge-distance

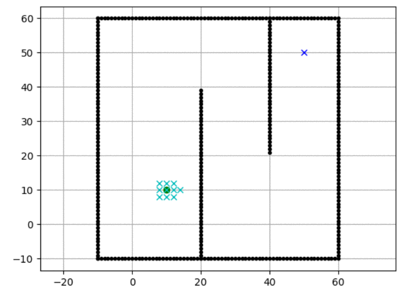
```
                seen.add(neighbor)
                frontier.put(neighbor)
                node2dist[neighbor] = node2dist[m] + 1
            else:
```

∞

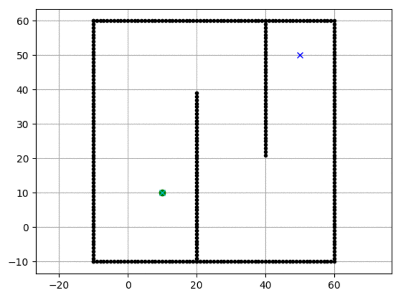
```
                node2dist[neighbor] = min(node2dist[neighbor], node2dist[m] + 1)
        i += 1
    if debug: print("%d) Q = " % i, list(frontier.queue))
    return search_order, node2dist

# Driver Code
print("Following is the Depth -First Search path")
print(dfs(graph, 's', debug=True))    # function calling
```

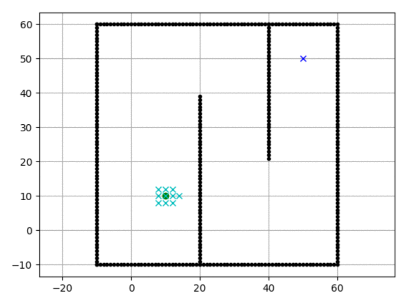
```
Following is the Depth -First Search path
0) Q =  ['s']; dists =  [0]
1) Q =  ['w', 'r']; dists =  [1, 1]
2) Q =  ['w', 'v']; dists =  [1, 2]
3) Q =  ['w']; dists =  [1]
4) Q =  ['t', 'x']; dists =  [2, 2]
5) Q =  ['t', 'y']; dists =  [2, 3]
6) Q =  ['t']; dists =  [2]
7) Q =  ['u']; dists =  [3]
8) Q =  []
(['s', 'r', 'v', 'w', 'x', 'y', 't', 'u'], {'s': 0, 'w': 1, 'r': 1, 'v': 2, 't': 2, 'x': 2,
```

2.0.5 Search order in BFS vs DFS vs Dijkstra

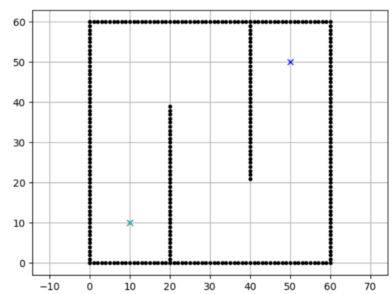


Breadth first search vs Depth first search





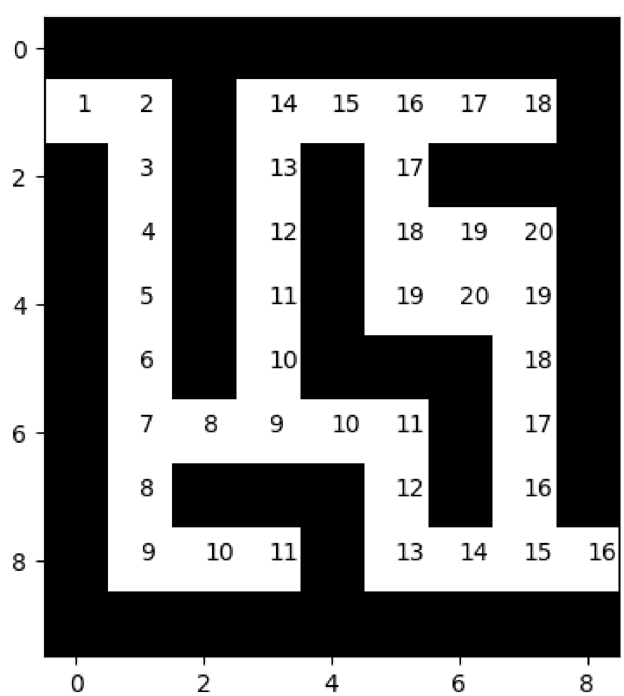
Breadth first search vs Dijkstra



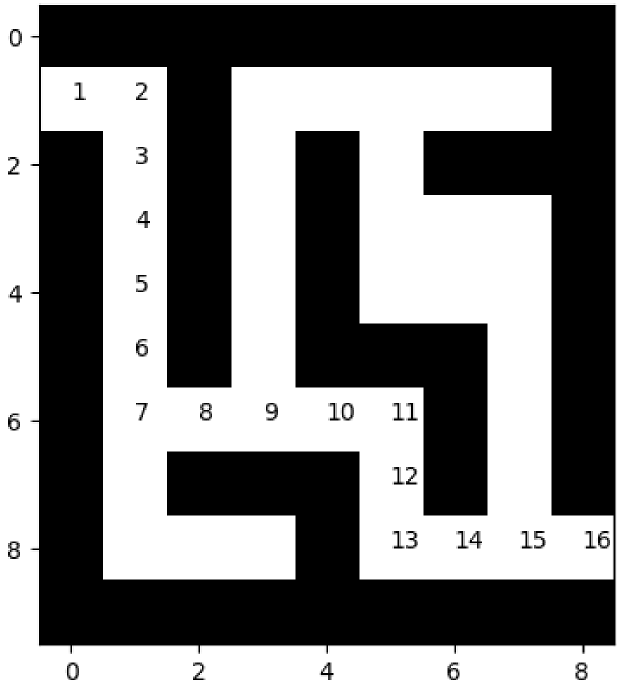
2.0.6 Converting a maze search to a graph search

```
import matplotlib.pyplot as plt
import numpy as np
maze_str = \
"""
+++++++++
  +     +
+ + + +++
+ + +   +
+ + +   +
+ + +++ +
+     + +
+ +++ + +
+   +
+++++++++
"""

class Maze:
    def __init__(self, maze_str, freepath=' '):
        self.maze_lines = [l for l in maze_str.split("\n")
                           if len(l)]
        self.FREEPATH = freepath
        self.fig = None

    def get(self, node, default):
        (r, c) = node
        m_row = self.maze_lines[r]
        nbrs = []
        if c -1 >= 0 and m_row[c -1] == self.FREEPATH:
            nbrs.append((r, c -1))
        if c+1 < len(m_row) and m_row[c+1] == self.FREEPATH:
            nbrs.append((r, c+1))
        if r -1 >= 0 and self.maze_lines[r -1][c] == self.FREEPATH:
```

```
            nbrs.append((r -1, c))
        if r+1 < len(self.maze_lines) and self.maze_lines[r+1][c] == self.FREEPATH:
            nbrs.append((r+1, c))
        return nbrs if len(nbrs) else default
    init_plots = init_plots
    plot_maze = plot_maze
    plot_step = plot_step
    plot_path = plot_path
    animate_search_path = animate_search_path

import matplotlib.pyplot as plt
import matplotlib.animation as animation
import matplotlib as mpl
%matplotlib inline
mpl.rc('animation', html='jshtml')


maze = Maze(maze_str)
search_path, node2dist = bfs(maze, (1, 0)) # prints the order of search all the searched noc
maze.plot_maze()

maze.animate_search_path(search_path, node2dist)
```



```
def bfs_path(graph, start, goal):
    """
    Returns success and node2parent

    success: True if goal is found otherwise False
    node2parent: A dictionary that contains the nearest parent for node
    """
    seen = [start] # List for seen nodes.
    # Frontier is the boundary between seen and unseen
    frontier = Queue() # Frontier of unvisited nodes as FIFO
    node2parent = dict() # Keep track of nearest parent for each node (requires node to be h
    frontier.put(start)

    while not frontier.empty():          # Creating loop to visit each node
        m = frontier.get() # Get the oldest addition to frontier
        if m == goal:
            return True, node2parent

        for neighbor in graph.get(m, []):
            if neighbor not in seen:
                seen.append(neighbor)
                frontier.put(neighbor)
                node2parent[neighbor] = m
    return False, []

def backtrace_path(node2parent, start, goal):
    c = goal
    r_path = [c]
    parent = node2parent.get(c, None)
    while parent != start:
        r_path.append(parent)
        c = parent
        parent = node2parent.get(c, None) # Keep getting the parent until you reach the sta
        #print(parent)
    r_path.append(start)
    return reversed(r_path) # Reverses the path

maze = Maze(maze_str)
start = (1, 0)
goal = (8, 8)
success, node2parent = bfs_path(maze, (1, 0), (8, 8))
path = backtrace_path(node2parent, (1, 0), (8, 8))
#print(list(path))
maze.plot_path(path) # Draws all the searched nodes
plt.show()
#node2parent
```



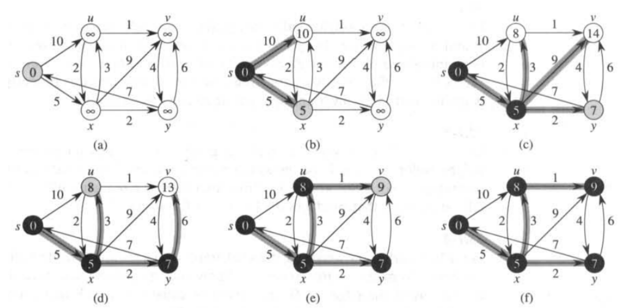


Figure 25.5 The execution of Dijkstra's algorithm. The source is the leftmost vertex. The shortest-path estimates are shown within the vertices, and shaded edges indicate predecessor values: if edge (u, v) is shaded, then $\pi[v] = u$. Black vertices are in the set S, and white vertices are in the priority queue $Q = V - S$. **(a)** The situation just before the first iteration of the **while** loop of lines 4–8. The shaded vertex has the minimum d value and is chosen as vertex u in line 5. **(b)–(f)** The situation after each successive iteration of the **while** loop. The shaded vertex in each part is chosen as vertex u in line 5 of the next iteration. The d and π values shown in part (f) are the final values.

Dijkstra algorithm

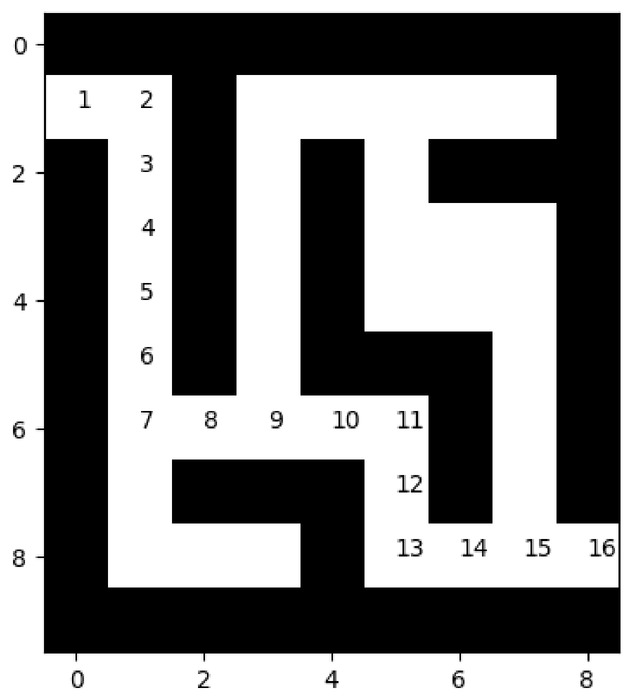
2.0.7 PriorityQueue

PriorityQueue returns the smallest (or the largest) item in the queue faster than other data structures

```
import itertools
```

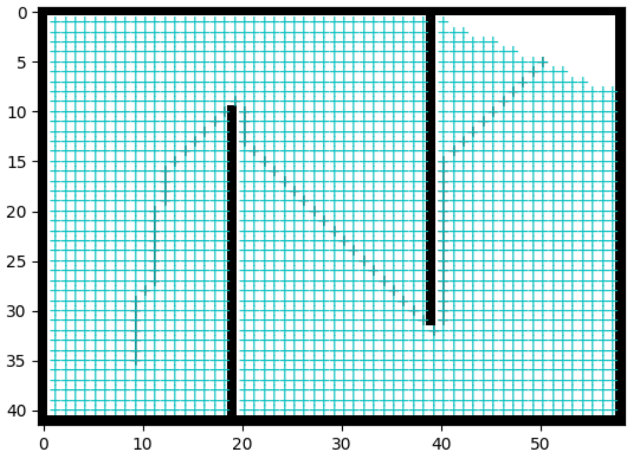
```
class MazeD(Maze):
    def get(self, node, default):
        nbrs = Maze.get(self, node, default)
        return zip(nbrs, itertools.repeat(1))

maze = MazeD(maze_str)
success, search_path, node2parent, node2dist = dijkstra(maze, (1, 0), (8, 8))
print(success, node2parent)
if success:
    path = backtrace_path(node2parent, (1, 0), (8, 8))
    maze.plot_path(path) # Draws all the searched nodes
```

```
True {(1, 0): None, (1, 1): (1, 0), (2, 1): (1, 1), (3, 1): (2, 1), (4, 1): (3, 1), (5, 1):
```

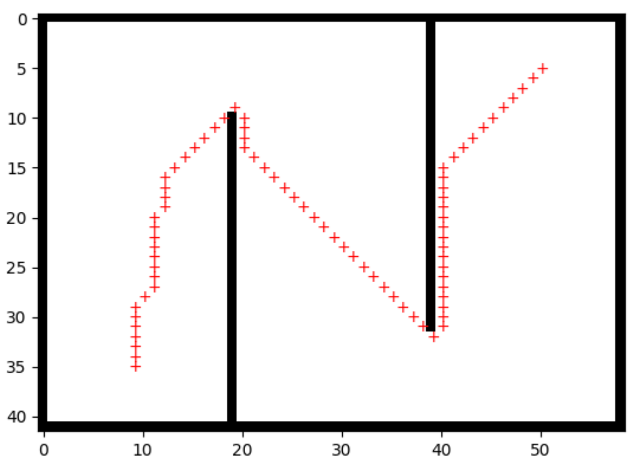



```
maze = Maze8(maze_str)
success, search_path, node2parent, node2dist = dijkstra(maze, start_pos, goal_pos)
#print(success, search_path)
assert success
anim = maze.animate(search_path)
path = backtrace_path(node2parent, start_pos, goal_pos)
#maze.init_plots(reinit=True)
path_plot = maze.plot_path(path, color='k') # Draws the traced shortest path
anim
```

```
/tmp/ipykernel_3055857/955263672.py:37: UserWarning: frames=<generator object batched at 0x7
  anim = animation.FuncAnimation(self.fig, self._plot_path,
```



```
path = backtrace_path(node2parent, (35, 9), (5, 50))
maze.init_plots(reinit=True)
maze.plot_path(path, color='r') # Draws the traced shortest path
plt.show()
```

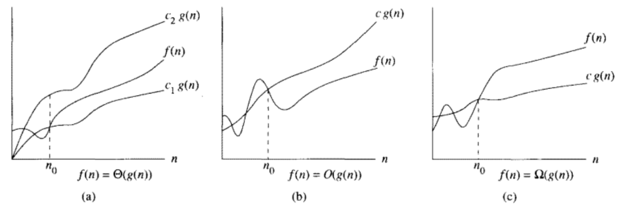


2.1 Computational complexity notation

Asymptotically tight bound $\Theta(g(n)) = \{f(n) :$ there exist positive constants $c_1, c_2,$ and n_o such that $0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0\}$

Asymptotic upper bound $O(g(n)) = \{f(n) :$ there exist positive constants $c_2,$ and n_o such that $0 \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0\}$

Asymptotic lower bound $\Sigma(g(n)) = \{f(n)$: there exist positive constants $c_1,$ and n_o such that $0 \leq c_1 g(n) \leq f(n)$ for all $n \geq n_0\}$



2.1.1 Computational complexity of BFS

```
# Write down the computational complexity of each line in big -O notation O()
# Assume the graph has |V| nodes and |E| edges
def bfs_barebones(graph, start):
    seen = {start} # Set for seen nodes (contains both frontier and dead states) # O(1)
    # Frontier is the boundary between seen and unseen (Also called the alive states)
    frontier = Queue() # Frontier of unvisited nodes as FIFO # O(1)
    frontier.put(start) # O(1)

    while not frontier.empty(): # Creating loop to visit each node # O(|V|)
        m = frontier.get() # Get the oldest addition to frontier # O(|V| * 1)

        for neighbor in graph.get(m, []): # O(|V| * |E|/|V|) = O(|E|)
            if neighbor not in seen: # O(|E| * 1)
                seen.add(neighbor) # O(|E| * 1)
                frontier.put(neighbor) # O(|E| * 1)
# The computational complexity of BFS is O(|E|). Some books write it as O(|V| + |E|)
# where O(|V|) is the cost of initializing states of different nodes
```

2.1.2 Computational complexity of Dijkstra

```
# Write down the computational complexity of each line in big -O notation O()
# Assume the graph has |V| nodes and |E| edges
def dijkstra_barebones(graph, start):
    seen = {start} # Set for seen nodes (contains both frontier and dead states) # O(1)
    # Frontier is the boundary between seen and unseen (Also called the alive states)
    frontier = PriorityQueue() # Frontier of unvisited nodes as PriorityQueue # O(1)
    frontier.put(PItem(0, start)) # O(1)
    node2dist = {start: 0} # Keep track of cost to arrive at each node # O(1)

    while not frontier.empty(): # Creating loop to visit each node
        dist_and_node = frontier.get() # Get the smallest dist node # O(|V| * log(|V|))
        m_dist = dist_and_node.dist
        m = dist_and_node.node
```

```
        for neighbor, edge_dist in graph.get(m, []): # O(|V| * |E|/|V|) = O(|E|)
            if neighbor not in seen: # O(|E| * 1)
                seen.add(neighbor) # O(|E| * 1)
                frontier.put(neighbor) # # O(|E| * log(1)) # for fibonacci heap
                node2dist[neighbor] = m_dist + edge_dist # O(1)
            elif node2dist[neighbor] > m_dist + edge_dist: # O(1)
                node2dist[neighbor] = m_dist + edge_dist # O(1)

# The computational complexity of Dijkstra is O(|V|log(|V|) + |E|) when implemented
# using a Fibonacci heap based PriorityQueue
```

2.2 PriorityQueue (Heaps Chapter 7 of Carmen's intro to algorithms)

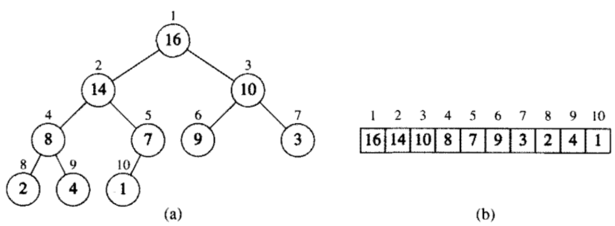


Figure 7.1 A heap viewed as **(a)** a binary tree and **(b)** an array. The number within the circle at each node in the tree is the value stored at that node. The number next to a node is the corresponding index in the array.

Heap property

1. $H[\text{Parent(i)}] \geq H[i]$
2. Parent(i) = ceil(i/2)
3. LeftChild(i) = 2i
4. RightRight(i) = 2i+1

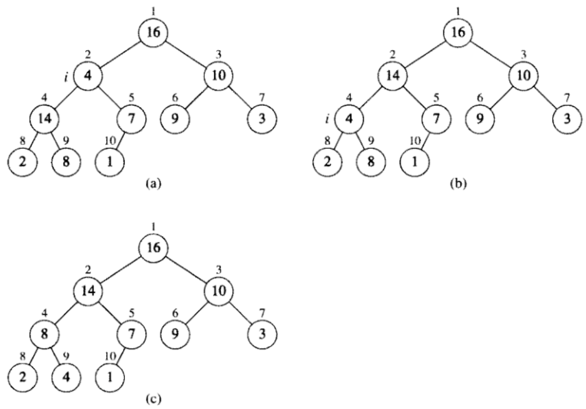


Figure 7.2 The action of HEAPIFY$(A, 2)$, where *heap-size*$[A] = 10$. **(a)** The initial configuration of the heap, with $A[2]$ at node $i = 2$ violating the heap property since it is not larger than both children. The heap property is restored for node 2 in **(b)** by exchanging $A[2]$ with $A[4]$, which destroys the heap property for node 4. The recursive call HEAPIFY$(A, 4)$ now sets $i = 4$. After swapping $A[4]$ with $A[9]$, as shown in **(c)**, node 4 is fixed up, and the recursive call HEAPIFY$(A, 9)$ yields no further change to the data structure.

Heapify

```
HEAPIFY(A, i)
  l ← LEFT(i)
  r ← RIGHT(i)
  if l ≤ heap-size[A] and A[l] > A[i]
     then largest ← l
     else largest ← i
  if r ≤ heap-size[A] and A[r] > A[largest]
     then largest ← r
  if largest ≠ i
     then exchange A[i] ↔ A[largest]
            HEAPIFY(A, largest)
```

Heapify pseudocode

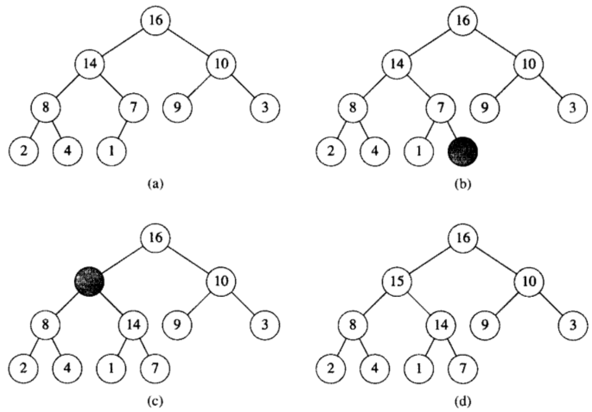


Figure 7.5 The operation of HEAP-INSERT. **(a)** The heap of Figure 7.4(a) before we insert a node with key 15. **(b)** A new leaf is added to the tree. **(c)** Values on the path from the new leaf to the root are copied down until a place for the key 15 is found. **(d)** The key 15 is inserted.

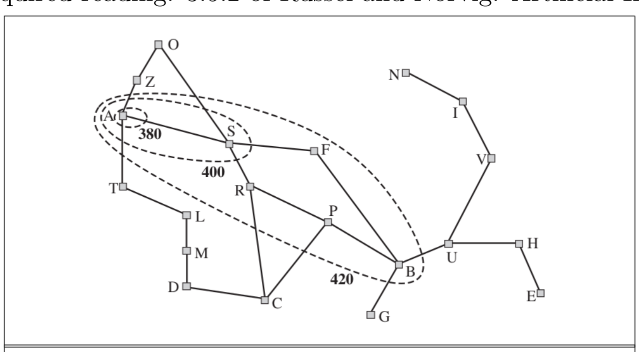
Heap Insert

Operation	find-min	delete-min	insert	decrease-key	meld
Binary[9]	$\Theta(1)$	$\Theta(\log n)$	$O(\log n)$	$O(\log n)$	$\Theta(n)$
Leftist	$\Theta(1)$	$\Theta(\log n)$	$\Theta(\log n)$	$O(\log n)$	$\Theta(\log n)$
Binomial[9][10]	$\Theta(1)$	$\Theta(\log n)$	$\Theta(1)$[a]	$\Theta(\log n)$	$O(\log n)$
Skew binomial[11]	$\Theta(1)$	$\Theta(\log n)$	$\Theta(1)$	$\Theta(\log n)$	$O(\log n)$[b]
Pairing[12]	$\Theta(1)$	$O(\log n)$[a]	$\Theta(1)$	$o(\log n)$[a][c]	$\Theta(1)$
Rank-pairing[15]	$\Theta(1)$	$O(\log n)$[a]	$\Theta(1)$	$\Theta(1)$[a]	$\Theta(1)$
Fibonacci[9][2]	$\Theta(1)$	$O(\log n)$[a]	$\Theta(1)$	$\Theta(1)$[a]	$\Theta(1)$
Strict Fibonacci[16]	$\Theta(1)$	$O(\log n)$	$\Theta(1)$	$\Theta(1)$	$\Theta(1)$
Brodal[17][d]	$\Theta(1)$	$O(\log n)$	$\Theta(1)$	$\Theta(1)$	$\Theta(1)$
2-3 heap[19]	$O(\log n)$	$O(\log n)$[a]	$O(\log n)$[a]	$\Theta(1)$	?

Heap runtimes

2.3 A-star (A*) algorithm

(Required reading: 3.5.2 of Russel and Norvig: Artificial Intelligence)



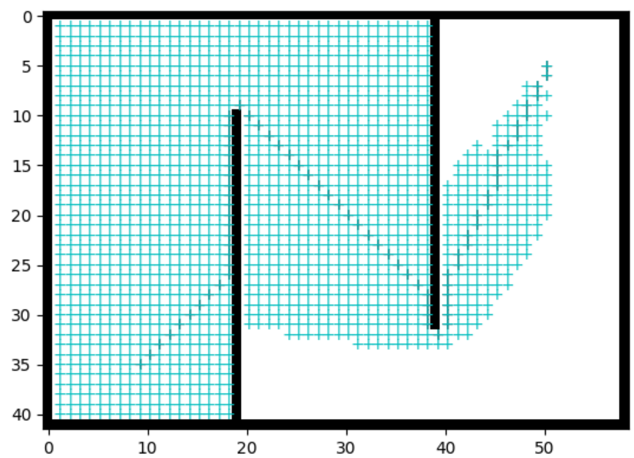

```
from hw2_solution import PriorityQueueUpdatable
```

```
import sys
def astar(graph, heuristic_dist_fn, start, goal, debug=False, debugf=sys.stdout):
    """
    edgecost: cost of traversing each edge

    Returns success and node2parent

    success: True if goal is found otherwise False
    node2parent: A dictionary that contains the nearest parent for node
    """
    seen = set([start]) # Set for seen nodes.
    # Frontier is the boundary between seen and unseen
    frontier = PriorityQueueUpdatable() # Frontier of unvisited nodes as a Priority Queue
    node2parent = {start : None} # Keep track of nearest parent for each node (requires node
    hfn = heuristic_dist_fn # make the name shorter
    node2dist = {start: 0  } # Keep track of cost to arrive at each node
    search_order = []
    frontier.put(PItem(0 + hfn(start, goal), start)) #   < - - - - - - - - - - - - - - Differe

    if debug: debugf.write("goal = "  + str(goal) + '\n')
    i = 0
    while not frontier.empty():          # Creating loop to visit each node
        dist_m = frontier.get() # Get the smallest addition to the frontier
        if debug: debugf.write("%d) Q = " % i + str(list(frontier.queue)) + '\n')
        if debug: debugf.write("%d) node = " % i + str(dist_m) + '\n')
        #if debug: print("dists = " , [node2dist[n.node] for n in frontier.queue])
        m = dist_m.node
        m_dist = node2dist[m]
        search_order.append(m)
        if goal is not None and m == goal:
            return True, search_order, node2parent, node2dist

        for neighbor, edge_cost in graph.get(m, []):
            old_dist = node2dist.get(neighbor, float("inf"))
            new_dist = edge_cost +  m_dist
            if neighbor not in seen:
                seen.add(neighbor)
                frontier.put(PItem(new_dist +  hfn(neighbor, goal), neighbor)) # < - - - - -
                node2parent[neighbor] = m
                node2dist[neighbor] = new_dist
            elif new_dist < old_dist:
                node2parent[neighbor] = m
                node2dist[neighbor] = new_dist
                # ideally you would update the dist of this item in the priority queue
                # as well. But python priority queue does not support fast updates
                # - - - - - - - - - - - - - - Different from dijkstra - - - - - - - - - - - -
```

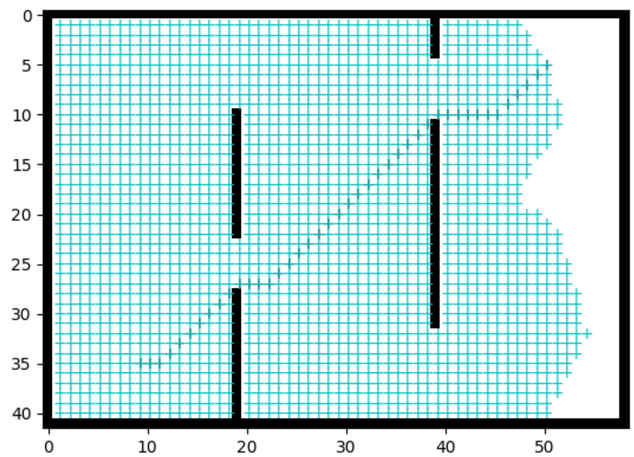
```
                old_item = PItem(old_dist + hfn(neighbor, goal), neighbor)
                if old_item in frontier:
                    frontier.replace(
                        old_item,
                        PItem(new_dist + hfn(neighbor, goal), neighbor))
        i += 1
    if goal is not None:
        return False, [], {}, node2dist
    else:
        return True, search_order, node2parent, node2dist
```

```
import math
from functools import partial

def euclidean_heurist_dist(node, goal, scale=1):
    x_n, y_n = node
    x_g, y_g = goal
    return scale*math.sqrt((x_n -x_g)**2 + (y_n - y_g)**2)
```

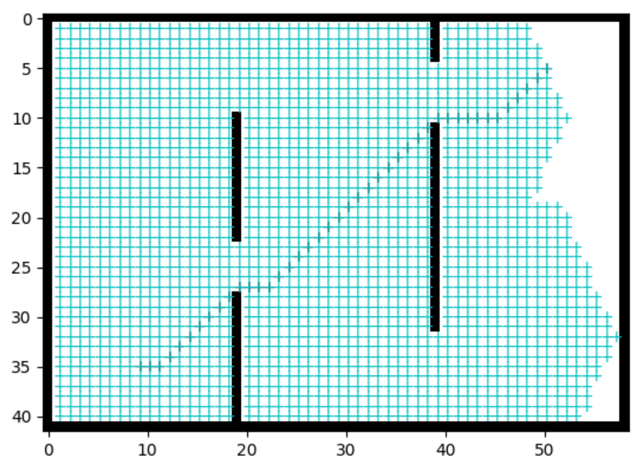
```
/tmp/ipykernel_3055857/955263672.py:37: UserWarning: frames=<generator object batched at 0x7
  anim = animation.FuncAnimation(self.fig, self._plot_path,
```




```
/tmp/ipykernel_3055857/955263672.py:37: UserWarning: frames=<generator object batched at 0x7
  anim = animation.FuncAnimation(self.fig, self._plot_path,
```

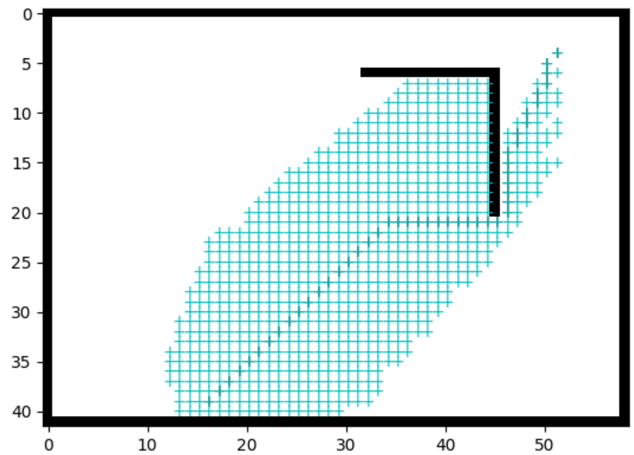


```
/tmp/ipykernel_3055857/955263672.py:37: UserWarning: frames=<generator object batched at 0x7
  anim = animation.FuncAnimation(self.fig, self._plot_path,
```



```
maze = Maze8(maze_str)
success, search_path, node2parent, node2dist = astar(
    maze, partial(euclidean_heurist_dist, scale=1),
    start_pos, goal_pos)
#print(success, search_path)
assert success
anim = maze.animate(search_path, batch_size=20)
path = backtrace_path(node2parent, start_pos, goal_pos)
#maze.init_plots(reinit=True)
path_plot = maze.plot_path(path, color='k') # Draws the traced shortest path
anim
```

```
/tmp/ipykernel_3055857/955263672.py:37: UserWarning: frames=<generator object batched at 0x7
  anim = animation.FuncAnimation(self.fig, self._plot_path,
```



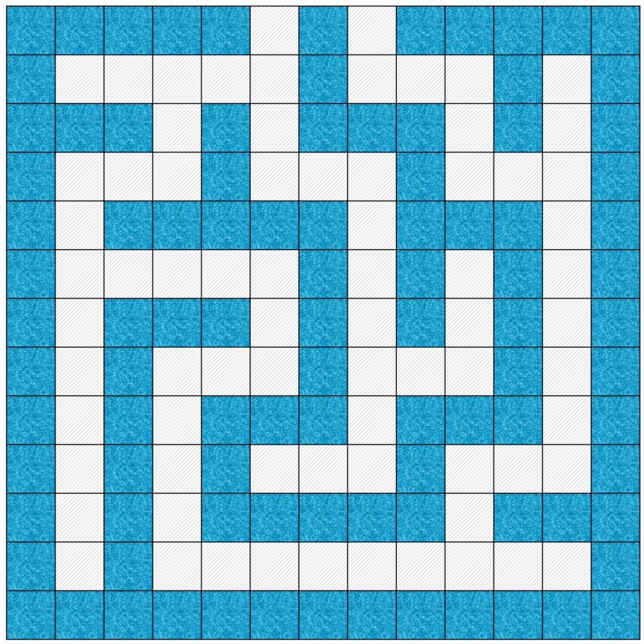
2.3.1 Admisibility and Consistency of heuristic function

1. An admissible heuristic is one that never overestimates the cost to reach the goal.

2. A heuristic h(n) is consistent if it satisfies the triangle inequality:

$h(\mathbf{s}_t) \leq c(\mathbf{s}_t, \mathbf{s}_{t+1}) + h(\mathbf{s}_{t+1})$.

2.4 Converting maze to grid to graph



2.5 Other ways of converting a maze into a graph

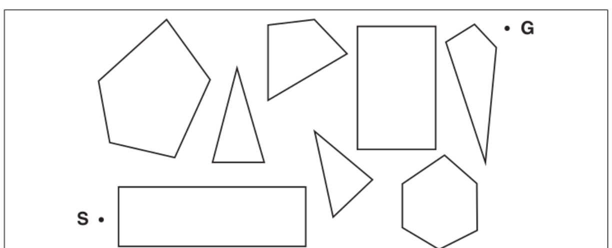


Figure 3.31 A scene with polygonal obstacles. S and G are the start and goal states.

2.6 Rapidly exploring random trees